Project Proposal
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For my final project, I would like to test new uncertainty design approaches against existing well-researched designs and measure the difference in users' accuracy, reliance, confidence, and trust. This is related to my literature review topic which was on LLM uncertainty quantification and uncertainty communication. I am interested particularly in uncertainty communication and hope to write a paper that is publishable in a HCI-related publication.

Within LLM uncertainty communication, design approaches with research activity include verbal hedging ("I'm not sure, but…"), numerical confidence indicators, visual confidence indicators, response modulation, abstention, and multiple-alternatives. However, I believe there are new design systems that would make worthy alternatives. Specifically, I would like to test the following new uncertainty communication designs. These would be tested against control (no uncertainty designs).

- **Communicate uncertainty by verbal hedging + providing web sources.** This is a more hybrid approach. When an LLM is uncertain, they should verbally express that uncertainty but also provide sources that users can then use to verify information for themselves.
- **Communicate uncertainty by having LLM actively verify claims.** Instead of simple verbal hedging, LLMs should say something like "However, you should double check [specific claim], because this is something I'm not entirely sure about."
- **Communicate uncertainty by giving data from other LLMs.** LLMs might say something like this within their responses: "Not all models agree on this though." or "3 out of 5 models agree on this."

[Note: still finalizing exactly what designs to test. Also, might make sense to only test 1 or 2 of these designs for the scope of this class.]

These designs will be tested against control and the main measurements will be:
- Accuracy: did the participant get the correct answer? (behavioral)
- Agreement: did the participant submit the same answer as LLM? (measure reliance)
- Confidence: how confident is the participant about the final answer? (this is self-reported)
- Trust: do you believe the LLM response is accurate? (this is self-reported)

The reason to measure all of these is that it gives a more full picture and makes for a more interesting research story. I would recruit participants on Mechanical Turk and/or Prolific. I will aim for 10-20 participants in each design group, though I might follow up with significantly more sample size afterwards; a small monetary incentive is likely necessary. I will not be 'building' any software for this project. Instead, participants will essentially partake in a survey where I show them the LLM questions and responses with the various designs directly in the survey.

Each participant will get approximately 8 LLM Q&As (4 where LLM is correct and 4 where LLM is incorrect). They will all get the "same" LLM Q&As except the only difference will be uncertainty design. Questions will be on topics that most people can't answer on their own so that users are forced to engage with the LLM. After each LLM Q&A, there will be a set of questions to measure the 4 parameters.

Disclosure:
I am also in Dr. Brian Smith's HCI Course (6178). In that class, I am also doing a final project/paper involving LLM uncertainty designs (with 2 other students). Currently, our research topic for that class is: *"How do various LLM uncertainty designs impact accuracy, reliability, confidence, and trust across different educational backgrounds?"* Essentially, we are testing the following uncertainty designs against each other but broken down by whether participants have gone to college or not: natural language hedging, numerical confidence score, visual confidence indicator, response modulation, and multiple alternatives.

There has been research on all of these design approaches, but it has all been on homogenous populations. My hypothesis is that there will be wide variance based on various demographic dimensions. It is important to study and acknowledge this or else some groups of LLM users might have over-reliance in high-stakes context leading to severe consequences. I also believe that there is no "one size fits all" design and likely LLM UX will have to communicate uncertainty with a hybrid approach taking into consideration the user, the context, and the question type.

For Dr. Smith's class, we are not coming up with and testing a new design. In that regard, there is no overlap with that class and 6156. However, there is definitely overlap in terms of knowledge sharing. For example, it is easy to say "accuracy, reliance, confidence, and trust", but it's harder to figure out how to best measure that within a survey. Whatever research and learnings I have regarding this, I will be able to apply to both Dr. Smith's class and 6156. Furthermore, my goal is to try to publish a research paper not too long after this semester ends; both Dr. Kaiser and Dr. Smith is aware of this. I can definitely see a scenario where I take the research findings from both classes and combine it into one paper. I am not sure if I will or if it makes sense, but I can envision it so want to make this disclaimer up front. Specifically, I can imagine this kind of research story:

- I test all the uncertainty designs based on educational background and get interesting findings/learnings (related to Dr. Smith's class).
- I analyze those findings and turn it into new uncertainty designs. (this is related to Dr. Kaiser's class). Obviously, I haven't actually run any tests so I don't know exactly how this story will flow, but it's possible that maybe I can use the results from this class's study and fit it in somehow. Most likely, I will have to at least add a lot to it somehow.
- Then in the paper, I can basically say something like this: I ran this test and did this analysis. Based on this analysis, I came up with new designs. I ran a test on these new designs and these are my findings which are promising for the future of uncertainty designs.